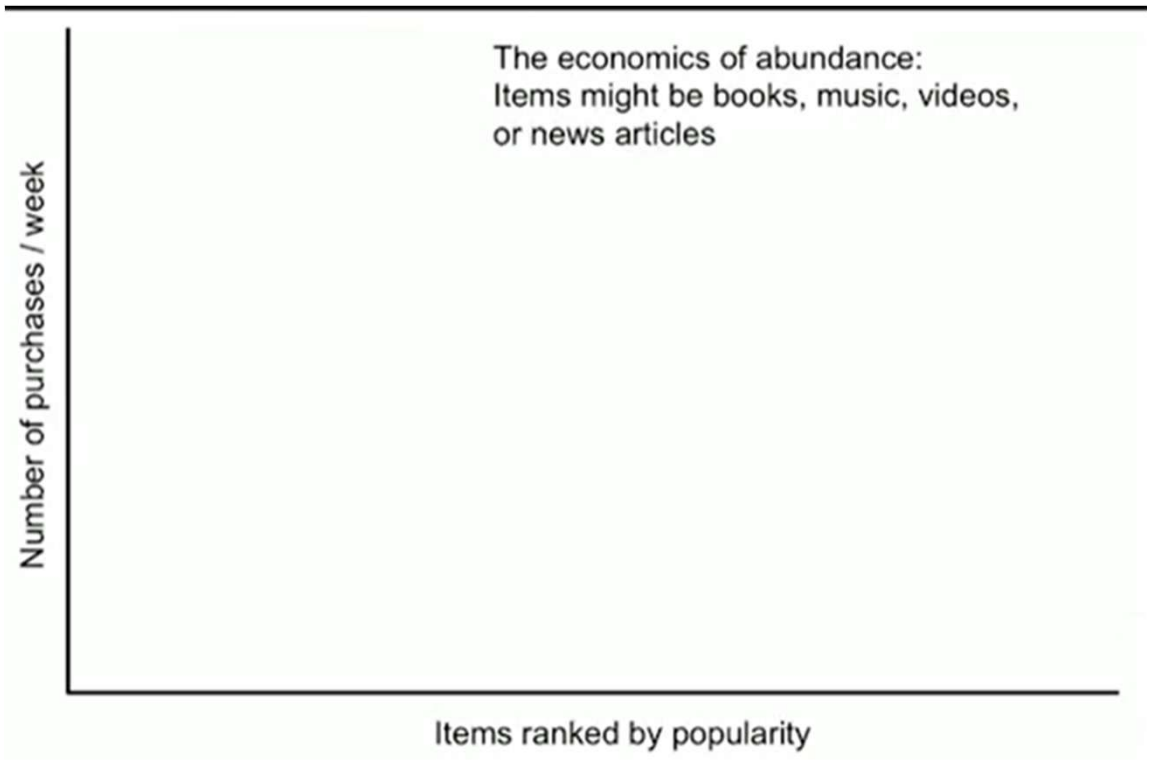
Recommender System

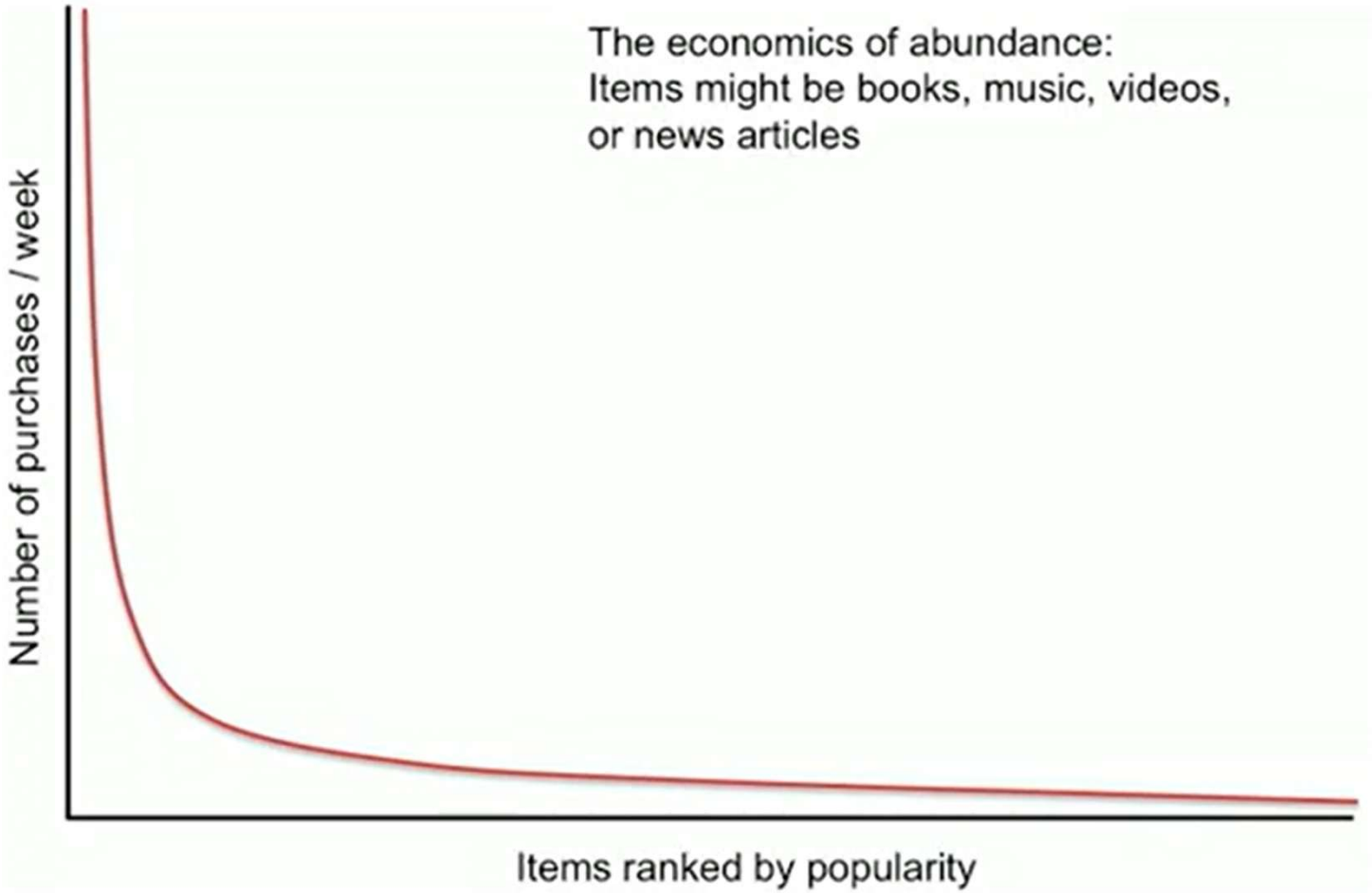
Over view –Recommender System

From scarcity to Abundance

- Shelf space is a scarce commodity for traditional retailers
  - Also: TV networks, movie theaters,…
- The web enables near-zero-cost dissemination of information about products
  - From scarcity to abundance
  - Gives rise to the “Long Tail” phenomenon

Long Tail


The economics of abundance:
Items might be books, music, videos,
or news articles
Number of purchases / week
Items ranked by popularity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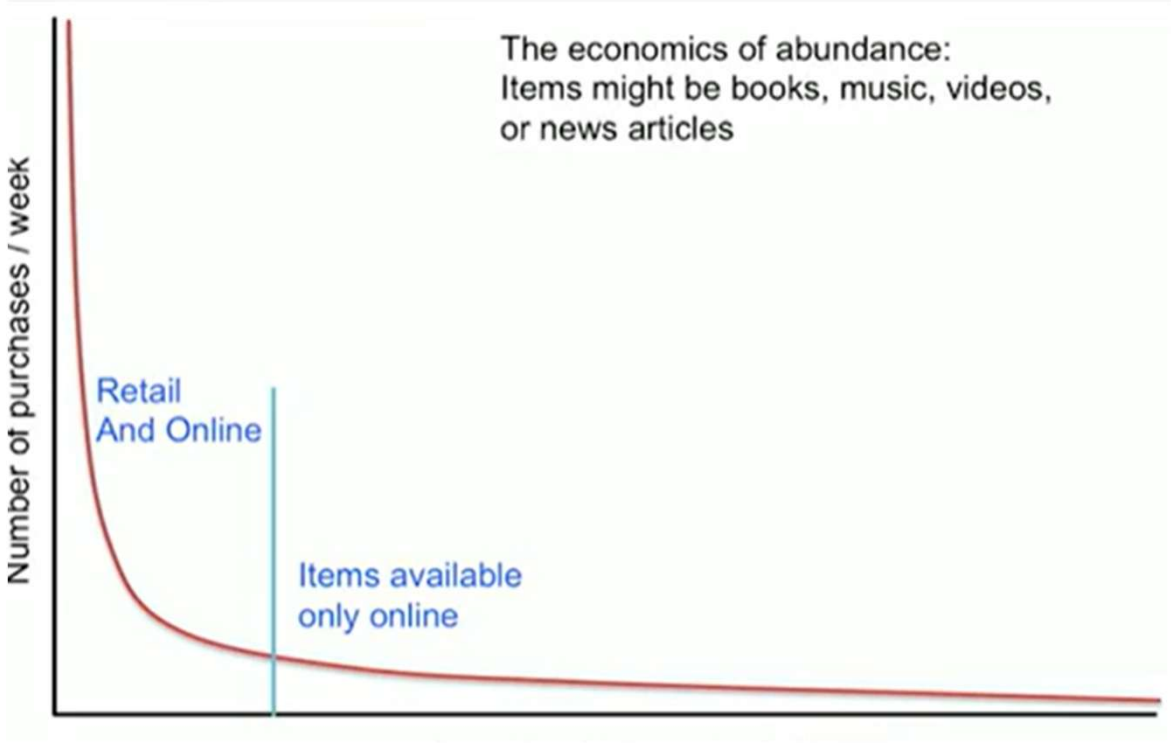
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Retail
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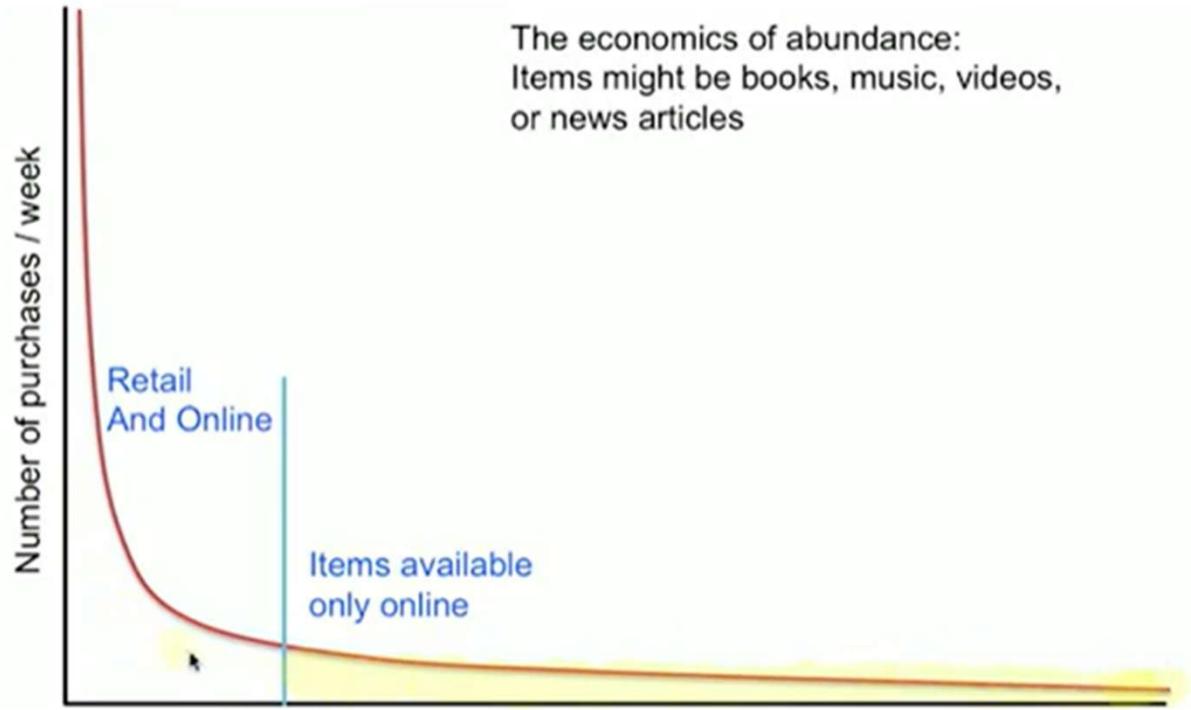
Items available
only online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- More choice necessitates better filters
  - Recommendation engines
  - How **Into Thin Air** made **Touching the Void** a bestseller (http://www.wired.com/wired/archive/12.10/tail.html)
- Examples
  - Books, movies, music, news articles
  - People (friend recommendations on Facebook, LinkedIn, and Twitter)

Types of Recommender Systems

- Editorial and hand curated
  - List of favorites
  - Lists of “essential” items
- Simple aggregates
  - Top 10, Most Popular, Recent Uploads
- Tailored to individual users
  - Amazon, Netflix, Pandora …
  - Our focus here

Formal Model

- C = set of Customers
- S = set of Items

- Utility function u: $C \times S \rightarrow R$
  - R = set of ratings
  - R is a totally ordered set
  - e.g., **0-5** stars, real number in **[0,1]**

Utility Matrix

	Avatar	LOTR	Matrix	Pirates
Alice	1		0.2	
Bob		0.5		0.3
Carol	0.2		1	
David				0.4

Key Problems

(1) Gathering "known" ratings for matrix

- How to collect the data in the utility matrix

(2) Extrapolate unknown ratings from the known ones

- Mainly interested in high unknown ratings
  - We are not interested in knowing what you don't like but what you like

(3) Evaluating extrapolation methods

- How to measure success/performance of recommendation methods

Gather rating

- Explicit
  - Ask people to rate items
  - Doesn't scale: only a small fraction of users leave ratings and reviews
- Implicit
  - Learn ratings from user actions
    - E.g., purchase implies high rating
  - What about low ratings?

Extrapolating Utilities

- Key problem: matrix ***U*** is **sparse**
  - Most people have not rated most items
  - Cold start:
    - New items have no ratings
    - New users have no history
- Three approaches to recommender system
  - **1)** Content-based
  - **2)** Collaborative
  - **3)** Latent factor based

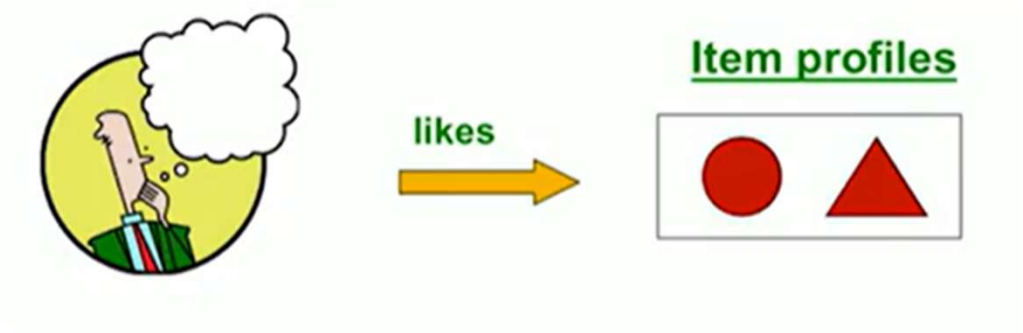
Content based Filtering

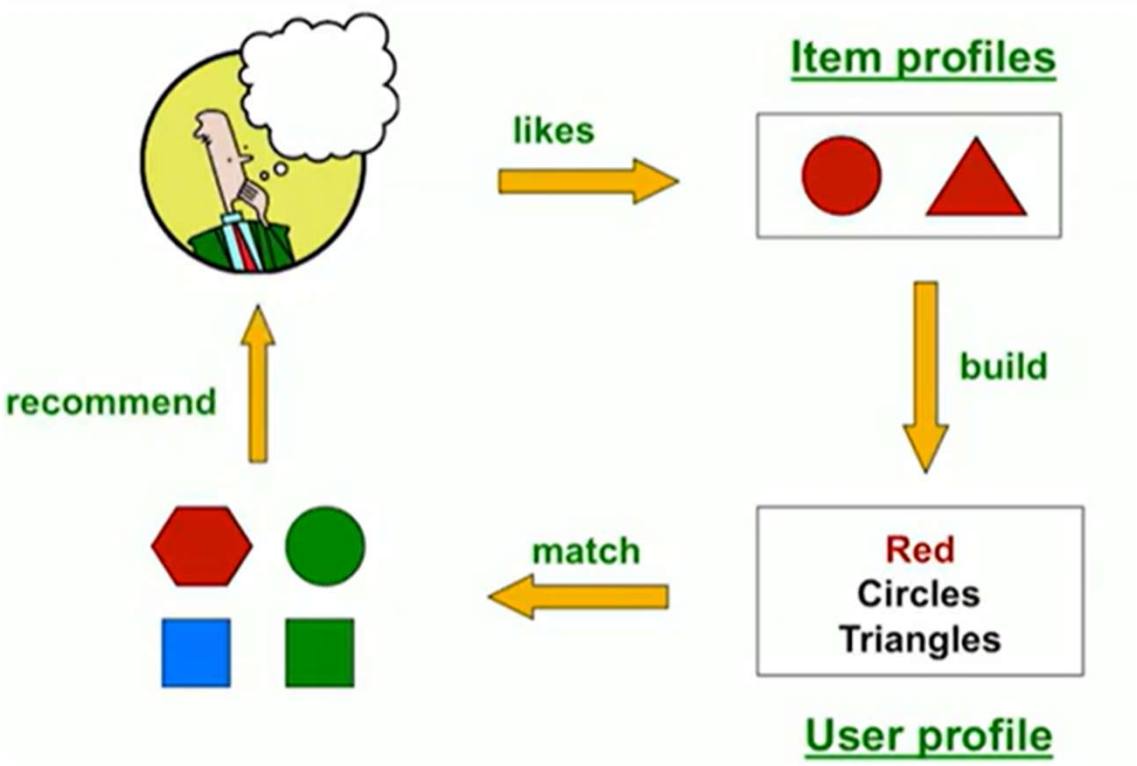
Main idea: Recommend items to customer ***x*** similar to previous items rated highly by ***x***

Examples:

- Movies
  - Same actor(s), director, genre, …
- Websites, blogs, news
  - Articles with "similar" content
- People
  - Recommend people with many common friends

Plan of action




Item profiles
likes
build
Red
Circles
Triangles
User profile
match
recommend

Item Profiles

- For each item, create an **item profile**
- Profile is a set of features
  - **Movies:** author, title, actor, director,…
  - **Images, videos:** metadata and tags
  - **People:** Set of friends
- Convenient to think of the item profile as a vector
  - One entry per feature (e.g., each actor, director,…
  - Vector might be boolean or real-valued

User profiles

- User has rated items with profiles $\mathbf{i}_1,\ldots,\mathbf{i}_n$
- Simple: (weighted) average of rated item profiles
- Variant: Normalize weights using average rating of user
- More sophisticated aggregations possible

Example1: Boolean Utility Matrix

- Items are movies, only feature is “Actor”
  - Item profile: vector with 0 or 1 for each Actor
- Suppose user x has watched 5 movies
  - 2 movies featuring actor A
  - 3 movies featuring actor B
- User profile = mean of item profiles
  - Feature A’s weight = 2/5 = 0.4
  - Feature B’s weight = 3/5 = 0.6

Example2: Star Ratings

- Same example, 1-5 star ratings
  - Actor A's movies rated 3 and 5
  - Actor B's movies rated 1, 2 and 4
- Useful step: Normalize ratings by subtracting user's mean rating (3)
  - Actor A's normalized ratings = 0, +2
    - Profile weight = (0 + 2)/2 = 1
  - Actor B's normalized ratings = -2, -1, +1
    - Profile weight = -2/3

Movie Title	Profile Vector	Action	Comedy
Die Hard	`[1, 0, 0]`	1	0
Matrix	`[1, 0, 1]`	1	0
Step Brothers	`[0, 1, 0]`	0	1
Galaxy Quest	`[1, 1, 1]`	1	1

2. What a "Weighted User Profile" looks like

For a user profile, we calculate the **mean (average)** of the item profiles they have watched (or liked). This vector doesn't use 0s and 1s; it uses **weights** between 0 and 1.

User History: Let's say User X has watched `Matrix` `[1, 0, 1]` and `Die Hard` `[1, 0, 0]`.

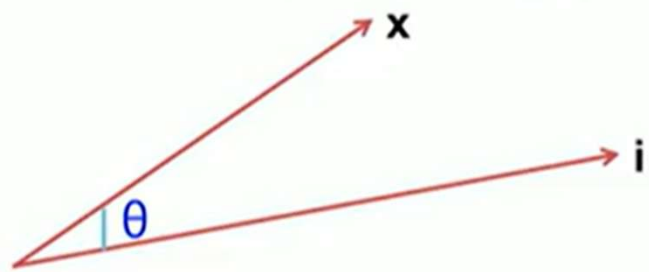
To calculate the User Profile, we find the average weight for each feature:

- **Action Weight:** (1 + 1) / 2 = **1.0**
- **Comedy Weight:** (0 + 0) / 2 = **0.0**
- **Sci-Fi Weight:** (1 + 0) / 2 = **0.5**

User Profile Vector: `[1.0, 0.0, 0.5]`

Making Predictions

- User profile **x**, Item profile **i**
- Estimate $U(\mathbf{x},\mathbf{i}) = \cos(\theta) = (\mathbf{x} \cdot \mathbf{i})/(|\mathbf{x}|\,|\mathbf{i}|)$



Technically, the cosine distance is actually the angle θ
And the cosine similarity is the angle $180-\theta$

For convenience, we use $\cos(\theta)$ as our similarity measure and call it the "cosine similarity" in this context.

Pros-Content Based

- No need for data on other users
- Able to recommend to users with unique tastes
- Able to recommend new & unpopular items
  - No first-rater problem
- Explanations for recommended items
  - Content features that caused an item to be recommended

Cons: Content Based

- Finding the appropriate features is hard
  - E.g., images, movies, music
- Overspecialization
  - Never recommends items outside user's content profile
  - People might have multiple interests
  - Unable to exploit quality judgments of other users

Collaborating Filtering

Collaborating Filtering

- Consider user ***x***
- Find set ***N*** of other users whose ratings are "**similar**" to ***x***'s ratings
- Estimate ***x***'s ratings based on ratings of users in ***N***

Similar Users

	HP1	HP2	HP3	TW	SW1	SW2	SW3
A	4			5	1		
B	5	5	4				
C				2	4	5	
D		3					3

- Consider users $\boldsymbol{x}$ and $\boldsymbol{y}$ with rating vectors $\boldsymbol{r_x}$ and $\boldsymbol{r_y}$
- We need a similarity metric sim($\boldsymbol{x}$, $\boldsymbol{y}$)
- Capture intuition that sim(A,B) > sim(A,C)

Option-1. Jaccard Similarity

	HP1	HP2	HP3	TW	SW1	SW2	SW3
A	4			5	1		
B	5	5	4				
C				2	4	5	
D		3					3

- sim(A,B) = $| r_A \cap r_B | / | r_A \cup r_B |$
- sim(A,B) = 1/5; sim(A,C) = 2/4
  - sim(A,B) < sim(A,C)
- Problem: Ignores rating values!

Option-2: Cosine Similarity

	HP1	HP2	HP3	TW	SW1	SW2	SW3
A	4			5	1		
B	5	5	4				
C				2	4	5	
D		3					3

- $sim(A,B) = \cos(r_A, r_B)$
- sim(A,B) = 0.38, sim(A,C) = 0.32
  - sim(A,B) > sim(A,C), but not by much

- **User A vector (r_A):** $[4, 0, 0, 5, 1, 0, 0]$
- **User B vector (r_B):** $[5, 5, 4, 0, 0, 0, 0]$

2. The Cosine Similarity Formula

The formula is the **Dot Product** divided by the **product of the magnitudes**:

$$\text{similarity}(A, B) = \cos(\theta) = \frac{\sum_{i=1}^{n} A_i B_i}{\sqrt{\sum_{i=1}^{n} A_i^2} \cdot \sqrt{\sum_{i=1}^{n} B_i^2}}$$

Multiply the corresponding ratings and add them up.

- $(4 \times 5) + (0 \times 5) + (0 \times 4) + (5 \times 0) + (1 \times 0) + (0 \times 0) + (0 \times 0)$
- $20 + 0 + 0 + 0 + 0 + 0 + 0 = \mathbf{20}$

B. Calculate the Magnitude (Length) of A:

Square each rating, sum them, and take the square root.

- $\sqrt{4^2 + 0^2 + 0^2 + 5^2 + 1^2} = \sqrt{16 + 25 + 1} = \sqrt{42} \approx \mathbf{6.48}$

C. Calculate the Magnitude (Length) of B:

- $\sqrt{5^2 + 5^2 + 4^2 + 0^2 + 0^2} = \sqrt{25 + 25 + 16} = \sqrt{66} \approx \mathbf{8.12}$

D. Final Division:

- $\text{sim}(A, B) = \frac{20}{6.48 \times 8.12} = \frac{20}{52.61} \approx \mathbf{0.38}$

Option 3: Centered Cosine

- Normalize ratings by subtracting row mean

	HP1	HP2	HP3	TW	SW1	SW2	SW3
A	4			5	1		
B	5	5	4				
C				2	4	5	
D		3					3

	HP1	HP2	HP3	TW	SW1	SW2	SW3
A	$2/3$			$5/3$	$-7/3$		
B	$1/3$	$1/3$	$-2/3$				
C				$-5/3$	$1/3$	$4/3$	
D		0					0

1. Calculate the Row Mean

First, find the average rating for each user, looking **only** at the movies they actually rated.

- **User A:** Rated $(4, 5, 1)$.
  - Mean = $\frac{4+5+1}{3} = \frac{10}{3} \approx \mathbf{3.33}$
- **User B:** Rated $(5, 5, 4)$.
  - Mean = $\frac{5+5+4}{3} = \frac{14}{3} \approx \mathbf{4.67}$
- **User C:** Rated $(2, 4, 5)$.
  - Mean = $\frac{2+4+5}{3} = \frac{11}{3} \approx \mathbf{3.67}$
- **User D:** Rated $(3, 3)$.
  - Mean = $\frac{3+3}{2} = \mathbf{3.0}$

Example: User A

- **HP1:** $4 - \frac{10}{3} = \frac{12}{3} - \frac{10}{3} = \mathbf{2/3}$
- **TW:** $5 - \frac{10}{3} = \frac{15}{3} - \frac{10}{3} = \mathbf{5/3}$
- **SW1:** $1 - \frac{10}{3} = \frac{3}{3} - \frac{10}{3} = \mathbf{-7/3}$

Example: User B

- **HP1:** $5 - \frac{14}{3} = \frac{15}{3} - \frac{14}{3} = \mathbf{1/3}$
- **HP3:** $4 - \frac{14}{3} = \frac{12}{3} - \frac{14}{3} = \mathbf{-2/3}$

Example: User D

- **HP2:** $3 - 3 = \mathbf{0}$
- **SW3:** $3 - 3 = \mathbf{0}$

Option 3: Centered Cosine similrity

	HP1	HP2	HP3	TW	SW1	SW2	SW3
A	$2/3$			$5/3$	$-7/3$		
B	$1/3$	$1/3$	$-2/3$				
C				$-5/3$	$1/3$	$4/3$	
D		0					0

- sim(A,B) = $\cos(r_A, r_B)$ = 0.09; sim(A,C) = -0.56
  - sim(A,B) > sim(A,C)
- Captures intuition better
  - Missing ratings treated as "average"
  - Handles "tough raters" and "easy raters"
- Also known as Pearson Correlation

Rating Predictions

- Let r_x be the vector of user x's ratings
- Let N be the set of k users most similar to x who have also rated item i
- Prediction for user x and item i

- Option 1: $r_{xi} = 1/k \sum_{y \in N} r_{yi}$
- Option 2: $r_{xi} = \sum_{y \in N} s_{xy} r_{yi} / \sum_{y \in N} s_{xy}$

  where $s_{xy} = \text{sim}(x,y)$

Item-item collaborating Filtering

- **So far: User-user collaborative filtering**
- **Another view: Item-item**
  - For item ***i***, find other similar items
  - Estimate rating for item ***i*** based on ratings for similar items
  - Can use same similarity metrics and prediction functions as in user-user model

$$r_{xi} = \frac{\sum_{j \in N(i;x)} s_{ij} \cdot r_{xj}}{\sum_{j \in N(i;x)} s_{ij}}$$

s_{ij}… similarity of items ***i*** and ***j***
r_{xj}…rating of user ***x*** on item ***j***
$N(i;x)$… set items rated by ***x*** similar to ***i***

Item-Item CF(|N|=2)

users

movies	1	2	3	4	5	6	7	8	9	10	11	12
1	1		3			5			5		4	
2			5	4			4			2	1	3
3	2	4		1	2		3		4	3	5	
4		2	4		5			4			2	
5			4	3	4	2					2	5
6	1		3		3			2			4	

- unknown rating

- rating between 1 to 5

Item-Item CF(|N|=2)

movies	1	2	3	4	5	6	7	8	9	10	11	12	sim(1,m)
1	1		3		?	5			5		4		1.00
2			5	4			4			2	1	3	-0.18
3	2	4		1	2		3		4	3	5		0.41
4		2	4		5			4			2		-0.10
5			4	3	4	2					2	5	-0.31
6	1		3		3			2			4		0.59

Neighbor selection:
Identify movies similar to movie **1**, **rated by user 5**

Here we use Pearson correlation as similarity:
1) Subtract mean rating m_i from each movie i
$m_1 = (1+3+5+5+4)/5 = 3.6$
row 1: *[-2.6, 0, -0.6, 0, 0, 1.4, 0, 0, 1.4, 0, 0.4, 0]*
2) Compute cosine similarities between rows

Item-Item CF(|N|=2)

users

movies	1	2	3	4	5	6	7	8	9	10	11	12	sim(1,m)
1	1		3		?	5			5		4		1.00
2			5	4			4			2	1	3	-0.18
3	2	4		1	2		3		4	3	5		0.41
4		2	4		5			4			2		-0.10
5			4	3	4	2					2	5	-0.31
6	1		3		3			2			4		0.59

Compute similarity weights:

$s_{13}=0.41$, $s_{16}=0.59$

Item-Item CF(|N|=2)

users

movies	1	2	3	4	5	6	7	8	9	10	11	12
1	1		3		2.6	5			5		4	
2			5	4			4			2	1	3
3	2	4		1	2		3		4	3	5	
4		2	4		5			4			2	
5			4	3	4	2					2	5
6	1		3		3			2			4	

Predict by taking weighted average:

r_{15} = (0.41*2 + 0.59*3) / (0.41+0.59) = 2.6

Item-item v user-user

- In theory, user-user and item-item are dual approaches
- In practice, item-item outperforms user-user in many use cases
- Items are "simpler" than users
  - Items belong to a small set of "genres", users have varied tastes
  - Item Similarity is more meaningful than User Similarity

- Popularity based
- Content based
- Collaborative based
- Hybrid

Popularity based

- Formula of company based

$$\mathrm{WR} = \frac{v}{v+m} \cdot R + \frac{m}{v+m} \cdot C$$

Variables

WR	weighted rating (dimensionless)
v	number of votes for the movie = (votes) (dimensionless)
m	minimum votes required to be listed in the Top 250 (currently 25000) (dimensionless)
R	average for the movie as a number from 0 to 10 (mean) = (Rating) (dimensionless)

Content based